

Day 23 Problem Set

Geometry Summer Credit | Modeling With Geometry

Name:	Date:	Period:
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Directions

For each modeling problem, write assumptions, choose a formula, calculate, and check whether the result makes sense.

Part A: Formula Choice and Assumptions

1.	A wall is 30 ft wide and 8 ft tall. One gallon of paint covers 320 square feet. How many gallons are needed for one coat?
2.	A rectangular garden is 22 ft by 14 ft. Edging costs \$3 per foot. Find the edging cost.
3.	A sandbox is 8 ft by 6 ft by 1.5 ft. Sand is sold by the cubic yard. Find the volume in cubic feet and decide what unit conversion is needed.
4.	A storage bin is 16 in by 12 in by 10 in. Does it hold at least 1800 cubic inches?
5.	A cylindrical water tank has radius 3 ft and height 8 ft. Find its volume in terms of pi.
6.	A design is scaled up by factor 2. Explain how the surface area and volume change.
7.	Error analysis: A student uses floor area to decide how much air fits in a room. Explain the modeling error.

Part B: Design Constraints

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| 8. | A box must hold at least 1000 cubic inches. Design A is 12 in by 9 in by 9 in. Does it meet the constraint? |
| 9. | Design B is a cylinder with radius 6 in and height 9 in. Find its volume in terms of pi. Does it clearly exceed 1000 cubic inches? Use pi approx 3.14 if helpful. |
| 10. | A package can use no more than 700 square inches of cardboard. A rectangular box is 14 in by 10 in by 8 in. Find surface area and decide if it meets the constraint. |
| 11. | A material has density 0.8 g/cm^3 . A solid has volume 600 cm^3 . Find its mass and decide if it is under a 500 g limit. |
| 12. | A scale model has surface area 45 square inches. The actual object has scale factor 4 from the model. Find actual surface area. |
| 13. | A scale model has volume 12 cubic inches. The actual object has scale factor 4 from the model. Find actual volume. |
| 14. | Describe one assumption you would make when modeling the amount of paint needed for a classroom. |

Day 23 Problem Set

Continued | Mixed Review and Challenge

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| 15. | Spiral review: Two similar solids have scale factor 3. The smaller volume is 25 cubic units. Find the larger volume. |
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| 16. | Spiral review: A cylinder has radius 4 and height 12. Find surface area in terms of π . |
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| 17. | Spiral review: A composite figure is a 20 by 12 rectangle with a 5 by 4 rectangle removed. Find the area. |
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| 18. | Application: A farmer wants to cover a rectangular greenhouse roof that is 40 ft by 18 ft. Plastic comes in rolls covering 500 square feet. How many rolls are needed? |
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| 19. | Challenge: A cylindrical can must hold at least 350 cm^3 . Compare $r=4$, $h=7$ and $r=5$, $h=5$. Which design holds more? Which may use less material? Show evidence. |
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